1. Introduction & Purpose

Team SPCE Robocon has been participating in Robocon India since 2010. The team secured 9th rank in 2019. The purpose of this report is to highlight the fabrication process of the two robots in compliance with the guidelines mentioned in the rulebook and design goals set by the team for each robot. We have utilized the following software to develop a technically sound design: CATIA V5®, FluidSIM®, ANSYS®, and EAGLE®.

2. Design Goals

2.1 Robust, compact, can sustain finite vibrations.
2.2 High-speed locomotion, pickup and placing.
2.3 Accurate catching mechanism adaptable to errors in throwing.
2.4 Highly accurate throwing and kicking.

3. Pass Robot

3.1 Chassis Design

(l×b×h: 600mm×600mm×450mm)

AISI 1018 Mild Steel (MS) square sections (0.5"×0.5"×1.2mm) are welded using argon welding to form a square shaped chassis. Medium Density Fiberboard (MDF) fixtures as shown in fig. 3.1 are designed and manufactured by laser cutting to ensure that the chassis rods are welded in the same plane and there is no thermal deformation of rods while welding. Chassis links are positioned in compliance with maximum deformation (0.1854mm) data, as obtained on ANSYS® as shown in fig. 3.2.

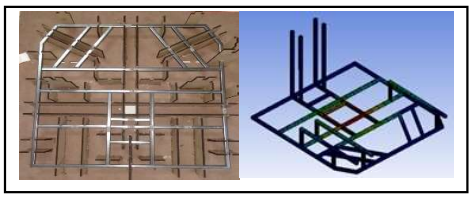


Fig. 3.1 MDF fixtures Fig. 3.2 ANSYS®

Three-wheel drive is preferred because of its easy maneuverability, simple control and the fact that wheels of three wheels chassis are always in a single plane.

3.2 Locomotion (Fig 3.3)

The load of the robot is uniformly distributed on three bidirectional Omni wheels (bearing type, ϕ100 mm). Three DC geared planetary motors (350 rpm, 12V) drive these wheels. A two-piece coupler acts as a coupling between motor and wheel. Aluminium coupling acts as a flange to support the wheel and another U-slotted MS coupler (ϕ10H7p6) press fits in a U-shaped motor shaft. Three cast nylon mounts manufactured using waterjet cutting and milling process are used for each motor-wheel assembly to avoid cantilever loading on motor body.

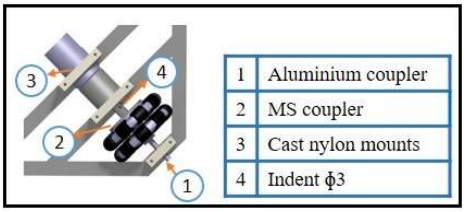


Fig. 3.3 Wheel Assembly

3.3 Try Ball Pickup (Fig. 3.4)

Pickup mechanism consists of two symmetrically placed wooden arms, designed considering the narrow space between two try balls in the rack and circular cross-section of try ball. Each arm has a piston (bore ϕ16mm, stroke length 25mm) that grips the ball when extended. Each arm has sponge attached, which help in gripping the ball properly. Whole assembly is rotated using DC geared Johnson motor (60 rpm-12V) coupled to the shaft using helical gears as shown in fig. 3.4. Gear ratio of 1:2 prevents failure of the motor shaft under excessive loading. This mechanism has only one degree of freedom that provides ease of operation. Cast Nylon mounts provide adequate support to the mechanism at appropriate place. Pistons are mounted exterior to the claw so that the arm in open position consumes minimum space. Another plus is that the claw grabs the ball in a single orientation that is beneficial for a precise throwing action. Ultrasonic sensors are used to attain optimum positioning of the robot during pickup of try ball.

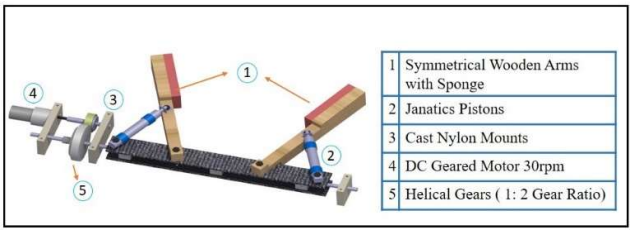


Fig. 3.4 Pickup Mechanism

2.5 Throwing Mechanism (Fig. 3.5)

Throwing mechanism design considers various factors like desired range, height of obstacles and optimum utilization of input pressure. Throwing plate has a hole (*ϕ130 mm*) matching the lower section of ball. Purpose of this slot is to position the try ball properly to assign an accurate trajectory. Use of two pistons (bore ϕ25mm, stroke length 100mm) is to increase the magnitude of impact force on try ball,

thus obtaining required projectile. Bearings (OD 19mm-ID8mm) for piston mounting are selected in order to sustain the piston force during throwing action. Aluminium mounts have counter bore which hold these bearings. The pneumatics circuit for the throwing mechanism was confirmed using FluidSIM® as shown in fig. 3.6

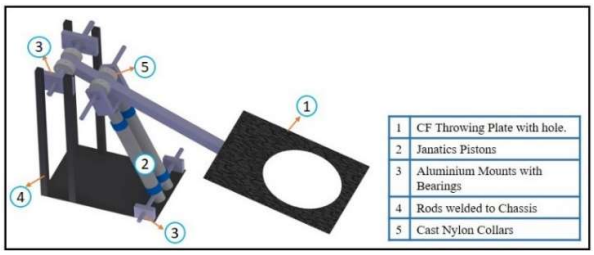


Fig. 3.5 Throwing Mechanism

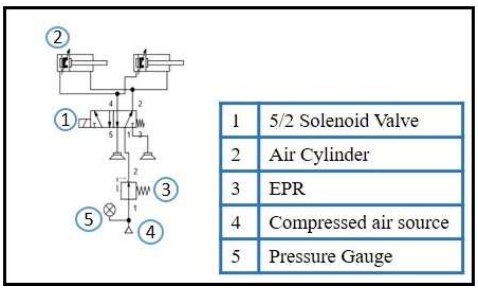


Fig. 3.6 FLUIDSIM

Calculations for Throwing

The optimum pressure required for throwing was determined by formulating some basic concepts of kinetics as shown below.

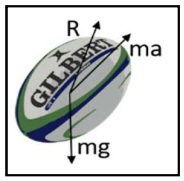


Fig 3.7 Free Body Diagram

Required range for the ball= 5m

Using projectile: $R = \frac{u^2 \sin 2\theta}{g}$

$\therefore u = \sqrt{\frac{5g}{\sin 40}}$ = 8.735m/s

Therefore, average acceleration of the ball at a distance of 0.455m from the hinge:

$$\omega_2^2 - \omega_1^2 = 2\alpha\theta$$

$$\therefore \frac{8.735^2}{0.455^2} - 0 = 2 \times \frac{a}{r} \times \frac{\pi}{180} 100$$

$\therefore$ a=48.04 m/s^2

R-mg=ma $\therefore$R=26.0325 N

Torque is constant

$\therefore \tau_{piston} = \tau_{ball}$ $\therefore Fr_1 = Rr_2$

$F = \frac{26.0325 \times 0.455}{0.07}$ = 169.21 N

Since the angle between the piston and throwing arm keeps on changing (while travelling in an arc)
Average value of force $F = F_p \sin\beta$
Considering average value of $\sin\beta = 0.8377$
$\therefore$ F_p = 202N (Approx.)
Bore diameter of piston, d=0.025m
$\therefore$ Pressure=4.11 bar

Use of Electronic Pressure Regulator (EPR)

The Electronic pressure regulator (SMC ITV 2050) supplies the required pressurized air to the pistons used for throwing, within a particular pressure range that can be controlled by providing electrical signals. EPR operates at 24V DC provided using DC-DC Boost Converter. Microcontroller provides control signal to EPR that controls the output pressure. Configuration of EPR is such that it provides high pressure (three to four bar) during extension of pistons whereas it lowers the air pressure (one bar) during retraction of pistons. This minimizes the impact of piston on robot's chassis while closing the pistons. This reduces vibrations effectively & minimizes bending of supporting links. In addition to this, EPR gives constant pressure output for a wide range of input air pressure, providing a constant trajectory of pass ball.

Carbon Fiber sheets are used extensively in throwing plate, pickup mechanism and placing mechanism owing to its promising strength to weight ratio as compared to SS sheets of similar dimensions.

3.5 Power Flow

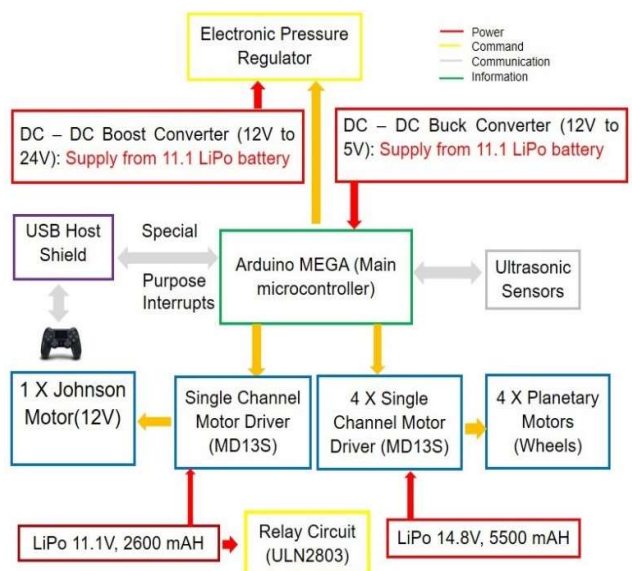


Fig 3.9 Power & Control Flow

4. Try Robot

4.1 Chassis Design

(l×b×h: 850mm×850mm×1100mm)

AISI 1018 MS square sections (0.75"×0.75"×1.9mm) were welded using argon welding to form square shaped chassis. Motive behind selecting 0.75" square section of MS was the increased section modulus that minimizes bending and keeps the stresses within

permissible limits, which makes it capable of sustaining huge forces generated by kicking action. MDF fixtures as shown in fig. 4.1, similar to pass robot were used in the welding process. Position of chassis links and trusses was finalized based on the maximum deformation (0.4461mm) data obtained from FEA on ANSYS® as shown in fig. 4.2. A special U-slot on chassis to provides a freeway for the sweep of kicking arm. Dimensions are optimized in accordance with maximum dimension constraints without compromising on the space requirements of each individual mechanism. Two MS rods used for kicking mechanism are bolted with its supporting links instead of welding for quick removal of whole mechanism to ensure easy maintenance

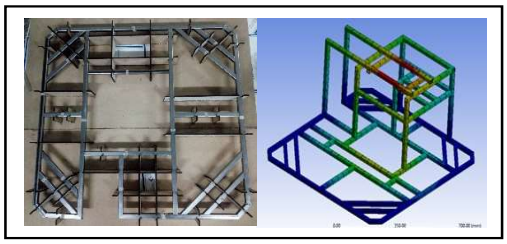


Fig. 4.1 MDF fixtures Fig. 4.2 ANSYS®

4.2 Locomotion

Total load on the robot is uniformly distributed over four bidirectional Omni wheels (Bearing type, ϕ100 mm). Four DC geared planetary motors (750 rpm-12V) drive these wheels. These high-speed motors help in quick movement of robot through different zones. The heavy weight of this robot warrants the use of four wheels to ensure permissible stress values in the wheel assemblies, particularly in the motor shafts. This also serves to avoid current overloading on each individual motor. Coupling between motor and wheels is done using a two piece MS coupler. First MS coupler acts as a flange to support the wheel & another D-slotted MS coupler (ϕ10H7p6) press fits in D-shaped motor shaft. Assembly of wheels and motor is similar to the PR robot.

4.2 Catching Mechanism: (Fig. 4.3)

Nylon net is framed in the shape of a funnel by introducing an upper and lower circular aluminum rings. The funnel shape is designed to absorb the momentum of incoming ball.

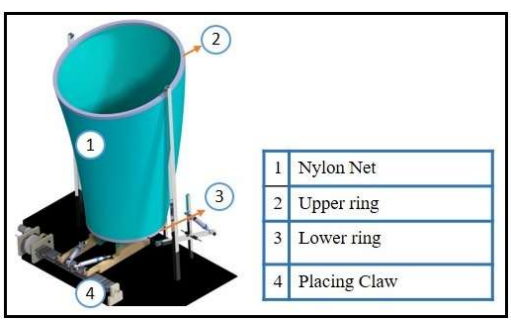


Fig. 4.3 Catching Mechanism

Catching area of the pass bot is therefore the area of the upper ring. The upper ring has a diameter of 500mm, while the lower ring has a diameter of 300mm.This shape gives a definite path to the motion of the ball and thus place it precisely at the desired position below the lower ring. The rings of funnel are manufactured using aluminum C-section of 1.2 mm thickness, bended using profile bending machine to get the desired circular shape.

3.5 Placing Mechanism (Fig. 4.4)

The received ball is grabbed in a claw having a set of symmetrical wooden arms. These arms are actuated using pistons (bore ϕ16mm, stroke length 25mm) that grip the ball when in retracted position. This assembly rotates by 180˚ to place the received ball in the try spot. This motion is provided by DC geared Johnson motor (30 rpm-12V), with helical gears of gear ratio 1:2 coupled to its shaft. Gears avoid direct loading on motor shaft. The claw has to grab the ball in any incoming position and it requires considerable spread. This condition is fulfilled by placing pistons inside claw's arms.

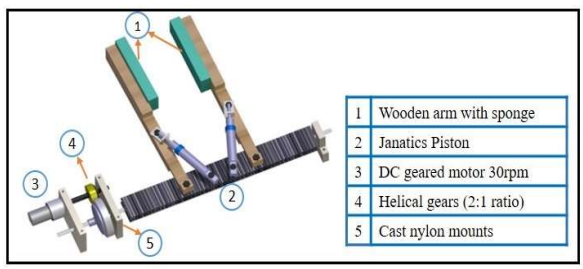


Fig. 4.4 Placing Mechanism

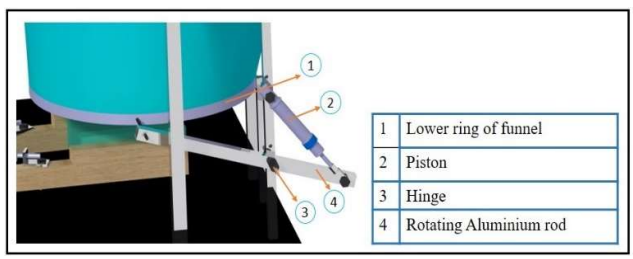


Fig. 4.5 Extra DOF to Lower Ring

Lower ring of funnel has one DOF enabled by a piston attached. It rotates about a hinge as shown in fig. 4.5. Initially, when the ball is being received, the piston is extended and the lower ring of the funnel is oriented in parallel to ground position. When the ball is to be placed, the piston is retracted thus the lower ring is oriented in perpendicular to ground position. This allows for unobstructed rotation of placing assembly while placing.

4.5 Kicking Mechanism

4.5.1 Concept

Kicking mechanism utilizes high-grade elastics in accordance to elasticity coefficient as calculated in section 4.5.5, owing to its high-energy conversion rate from potential energy to kinetic energy to obtain the needed kicking velocity. Kicking process involves

principle similar to catapult mechanism. The mechanism consists of a kicking arm, winding rope, elastics, power screw mechanism, wounding surface and pulling motor platform. One of the team member will place ball and tee in the arena manually. A LIDAR sensor is used to adjust the robot in the required position to obtain a precise kicking action.

4.5.2 Kicking Arm (Fig. 4.6)

The kicking arm is a pair of MS rods (0.5" × 0.5" ×1.2mm). It rotates about a hinge as shown in fig. 4.6. The elastics are attached to upper end of the arm. The lower extreme has a rubber slab placed on aluminum plate. This plate is inclined at an angle (45˚) in order to provide a proper angle of projection.

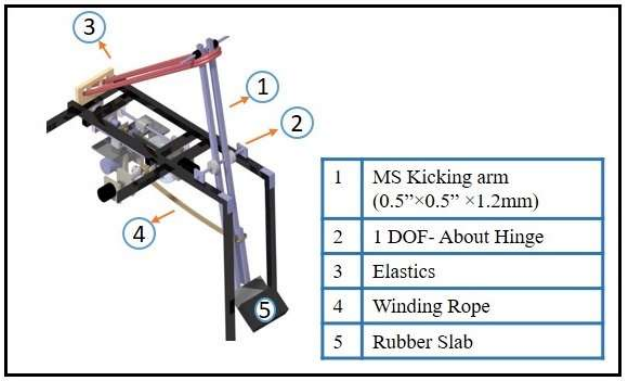


Fig. 4.6 Kicking Arm

Rigid rubber slab is used as the contact between ball and arm because of its good coefficient of restitution allowing maximum transfer of kinetic energy from arm to the ball.

4.5.3 Pulling action

A DC geared planetary motor (M1) (750 rpm-12V) is used to wound a string that pulls back the arm and creates tension in the elastic rope. Rope winds on a specially designed aluminium dumbbell shaped surface as shown in fig 4.7, which is coupled with gear1 (Nylocast-60 teeth). Coupling between motor shaft and gear2 (MS-15 teeth) is via a D-shaped MS coupler. The motor is placed on platform, which moves using linear bearings and power screw mechanism as shown in fig 4.8. Here a DC planetary geared motor (M2) (1000rpm-12V) rotates the power screw which is coupled with motor shaft using aluminum coupler. This rotation results in linear motion of SS C-section, which is bolted to the platform. It has a set of two copper nuts. As the shaft rotates, the copper nuts tends to move linearly. Two linear bearings are attached to platform, which traverse over two stationary SS shafts (ϕ8 mm) to avoid cantilever loading on power screw. In the wounding process, the motor M2 operates to establish proper meshing between gear 1 and 2, followed by rotation of motor M1, which eventually pulls the arm.

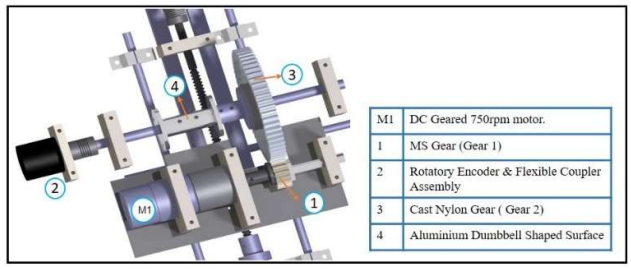


Fig. 4.7 Pulling Mechanism

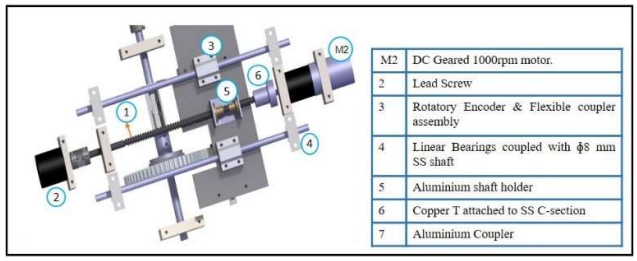


Fig. 4.8 Power Screw Mechanism

Encoder on gear1 controls the wounding of string and thus the amount of energy stored in elastics. Encoder on power screw mechanism helps in proper meshing of gears.

4.5.4 Release Process

It involves un-meshing of gears, thus eliminating the tension in the elastics. To execute this, the platform of motor M1 moves away from gear1 using motor M2 and its supporting mechanism. This assures speedy release of the arm that causes the desired kicking action.

4.5.5 Calculations of Elasticity coefficient of elastics

Symbol Conventions:
I = Moment of Inertia about pivot, ω = Angular Velocity about pivot, k = Elasticity constant
In the subscript: B = Ball, L=Leg
For elastic:-
Free length l_f = 28cm, Maximum stretched length l_s = 75cm, length just before the kick l_k = 54cm

Considering kicking zone 3 to be the launch point, the required distance and height are 8.7m and 2m respectively.
Applying equation of parabolic trajectory:

$$y = x\tan\theta - \frac{gx^2}{2u^2}(1+\tan^2(\theta))$$

and assuming θ=45˚, x=8.7, y=2;
u=10.522m/s (i.e. velocity just after the collision)
Now, through conservation of energy,
Work done by elastic = Rotational Kinetic Energy of the leg just before the kick

$$\frac{1}{2}k(l_s-l_f)^2-\frac{1}{2}k(l_s-l_k)^2=\frac{1}{2}I_L\omega_{L1}^2$$

Hence, $0.1533k = I_L\omega_{L1}^2$ --- Equation 1

I_L from CATIA Analysis is 0.097kgm^2

To find ω_{L1} :-

Applying conservation of angular momentum about pivot between the leg and the ball in the instant of collision:
$I_L\omega_{L1} = I_L\omega_{L2} + I_B\omega_B$ --- Equation 2;

Considering ball to be an ellipsoid and the distance between the ball and pivot(r) to be 0.42m (by measurement), $I_B = 0.0858 kgm^2$

Equation 2 has only two unknown quantities (ω_{L1} ω_{L2}) since $\omega_B = u/r$ is known.
(Although generally $\omega r\sin\theta = u$, here, $\theta = 90^\circ$ because collision between leg and ball is assumed head on)
Hence, ω_B=25.052rad/s

Second equation relating angular velocities of the ball and leg can be found through consideration of coefficient of restitution (e) = $\frac{Velocity\ of\ Separation}{Velocity\ of\ Approach}$
Velocity of separation= $\omega_B r - \omega_{L2} r$ and Velocity of approach = $\omega_{L1} r$
e = 0.8 (found by experimentation)
Putting appropriate values in Equation 1 and equation 2 for e, we get
ω_{L1} = 26.23rad/s and ω_{L2} = 4.069rad/s.

Substituting ω_{L1} back in Equation 1, we get
k = 435.34 N/m
This value would be appropriate under ideal conditions. Hence, considering all possible losses k = 870 N/m. (i.e. k_{calc}×2)

4.5.4 Damping of Excess Vibrations

Impact of the kick generates undesired vibrations, which need to be dissipated properly. Therefore, the rebound velocity of kicking arm has been minimized during its return swing. Following methods are designed for damping these vibrations:

1) Use of elastics to suppress extra rotation of arm. (Currently implemented on robot)
2) Set of compression springs
3) High Strength rubber bands

Safety Measure: Rollers attached to the sides of chassis of both robots ensures minimal damage to robot if it strikes the walls of arena. Emergency switch cuts off the supply to motors in case of accidental unexpected conditions.

4.6 Power Flow

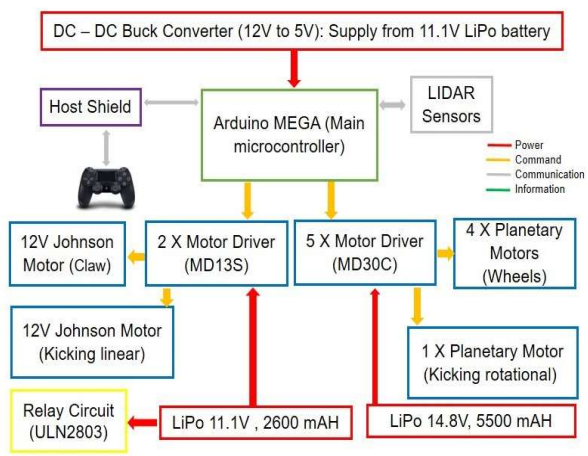
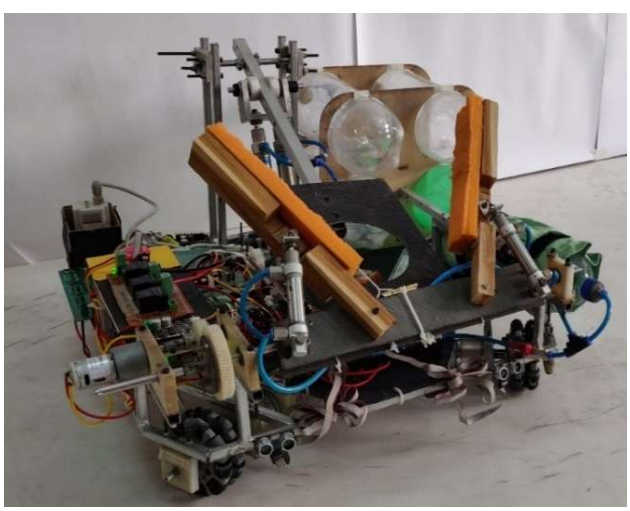


Fig. 5.9 Power & Control Flow

Current Status of Robots (As on 2/02/2020)



Pass Robot

Try Robot